SPEAKERS



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Ochrona kontenerów w środowisku multicloud

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Securing Kubernetes and container workloads across Azure, AWS, GCP and on-premises

TODAY'S SESSION

Agenda

01
Need for Container Security

Why containers expand the attack surface

02
KSPM

Kubernetes security posture management

03
Microsoft Defender for Cloud

The CNAPP platform and multicloud reach

04
Defender Capabilities

Posture, vulnerabilities, runtime protection

05
How It Works

Agentless signals, sensor and connectors

06
Demo

Defender for Containers in action

Containers changed how we build software — and how it gets attacked



Dynamic and ephemeral

Clusters autoscale and workloads come and go in seconds. Static inventories and periodic scans cannot keep up.



Shared kernel, large control plane

Many workloads share a node, and the Kubernetes API, RBAC and network policy become high-value targets.



Third-party supply chain

Images are built from public base layers and open-source packages, so a vulnerable layer ships everywhere.

The takeaway: protection has to follow the full lifecycle — build, registry, cluster configuration and runtime — in every cloud you run.

Risk appears at every stage of the container lifecycle

Build

RISK

Vulnerable base images and packages; secrets baked into layers

CONTROL

Image scanning before release

Registry

RISK

Stale images that were clean at push time but are now exploitable

CONTROL

Registry scans with daily re-scan

Cluster

RISK

Privileged pods, over-permissive RBAC, exposed API endpoints

CONTROL

Posture checks against best practice

Runtime

RISK

Reverse shells, cryptominers, privilege escalation, lateral movement

CONTROL

Behavioral threat detection

Static controls address the state of a container at or before deployment; once it is running, only runtime detection sees an attack in progress.

KSPM: continuously knowing how safely your clusters are configured

What is KSPM?

Kubernetes Security Posture Management continuously checks cluster and workload configuration against security best practices and compliance benchmarks, then helps you prioritize and fix the gaps.



Typical findings KSPM surfaces

Privileged containers and risky capabilities
Pods that can escape to the node

Over-permissive RBAC
Cluster-admin bindings and broad roles

Exposed services and open networks
Public endpoints, no network policies

Missing hardening controls
Resource limits, read-only filesystems, pod security

Vulnerable images in use
Known CVEs in running workloads

Defender CSPM delivers agentless container posture across Azure, AWS and GCP



Agentless discovery

Zero-footprint, API-based discovery of clusters, configurations and deployments.



Inventory and security explorer

Explore clusters, workloads, node pools, registries and images in one graph.



Agentless vulnerability assessment

Scans node pools and images, with near real-time scans of new images and daily refresh.



Attack path analysis

Contextual risk shows exploitable paths, so you fix what matters first.

Enable it

1. Turn on the Defender CSPM plan for the subscription or connector
2. Open Settings and enable Kubernetes API access and Registry access
3. For AWS and GCP, update the connector and redeploy its template

Microsoft Defender for Cloud: one CNAPP for code, cloud and runtime



Posture management

Defender CSPM: inventory, recommendations, compliance and attack path analysis.



Workload protection

Defender plans for containers, servers, databases, storage and more.



DevOps security

Code-to-cloud visibility from repositories and pipelines to running workloads.



Multicloud and hybrid

Azure, AWS, GCP and on-premises resources managed from a single portal.

Shared foundation: one security graph, one set of recommendations and alerts, and one place to manage every cloud.

One control plane for AKS, EKS, GKE and Arc-enabled Kubernetes

Environment	Onboarding path	How protection is delivered
Azure Kubernetes Service (AKS)	Native Azure resource; no connector needed	Azure-native integrations deploy the components; images in Azure Container Registry assessed
Amazon EKS	Onboard the AWS account with a connector	Agentless API access plus Arc-based provisioning of the Defender sensor
Google GKE	Onboard the GCP project with a connector	Agentless API access plus Arc-based provisioning of the Defender sensor
On-premises / other Kubernetes	Connect the cluster to Azure Arc	Arc-enabled Kubernetes with the Defender sensor deployed to the cluster

Support varies by plan and cloud; always check the current container support matrix before rollout.

Defender for Containers secures the supply chain and the running cluster together



Security posture management

Continuous monitoring of cloud APIs, Kubernetes APIs and workloads, with misconfiguration findings and mitigation guidance.

Agentless



Vulnerability assessment

Images in registries, running containers and Kubernetes nodes, covering OS and language packages with daily re-scans.

Agentless



Runtime threat protection

Detections for clusters, nodes and workloads, mapped to MITRE ATT&CK, with automated response options.

Defender sensor



Control plane threat detection

Detects suspicious activity against the Kubernetes API, with no components deployed in the cluster.

Agentless

Agentless vulnerability assessment from registry to running node

What gets scanned

Registry images

Scanned on push and re-scanned periodically

Running containers

Images currently deployed in clusters

Kubernetes nodes

Node pools in supported clusters

What you get

- **Zero configuration** No agent or scanner to deploy or maintain
- **Daily re-scans** Results refresh as new CVEs are published
- **OS and language packages** Coverage beyond the base image layer
- **Exploitability insights** Prioritize CVEs that can actually be exploited
- **Signed findings** Results are signed with a Microsoft certificate and linked to the image in the registry
- **Feeds the security graph** Vulnerabilities add context to attack paths and hunting

Runtime threat protection catches attacks that static controls miss

What it detects

- Malicious processes in containers
- Reverse shells and suspicious connections
- Cryptominer deployment
- Privilege escalation
- Lateral movement across the cluster
- Suspicious Kubernetes API activity

From signal to response



Collect

Sensor and control plane signals from every cluster

Detect

Microsoft threat intelligence and behavioral analytics

Map and alert

Alerts mapped to MITRE ATT&CK for context

Respond

Automated response and export to your SIEM or SOAR

AKS, EKS and GKE: near-parity coverage, with node scanning still in preview on AWS and GCP

Capability	Delivery	AKS	EKS	GKE
Agentless discovery and posture	Agentless	GA	GA	GA
Attack path analysis (Defender CSPM)	Agentless	GA	GA	GA
Registry image vulnerability assessment	Agentless	GA	GA	GA
Runtime container VA (registry-scan based)	Agentless	GA	GA	GA
Kubernetes node vulnerability assessment	Agentless	GA	Preview	Preview
Runtime workload threat detection	Sensor	GA	GA	GA
Binary drift detection and blocking	Sensor	GA	GA	GA
Gated deployment (Kubernetes 1.31+)	Sensor	GA	GA	GA

Linux workloads shown; Windows support differs. Source: Containers support matrix, Microsoft Learn. Always verify the current matrix.

How it works: most capabilities are agentless, runtime detection uses a sensor

Capability	Agentless	In-cluster component
Image vulnerability assessment	Yes	No
Kubernetes posture assessment	Yes	No
Runtime threat detection	No	Yes
Control plane threat detection	Yes	No



Registry connection

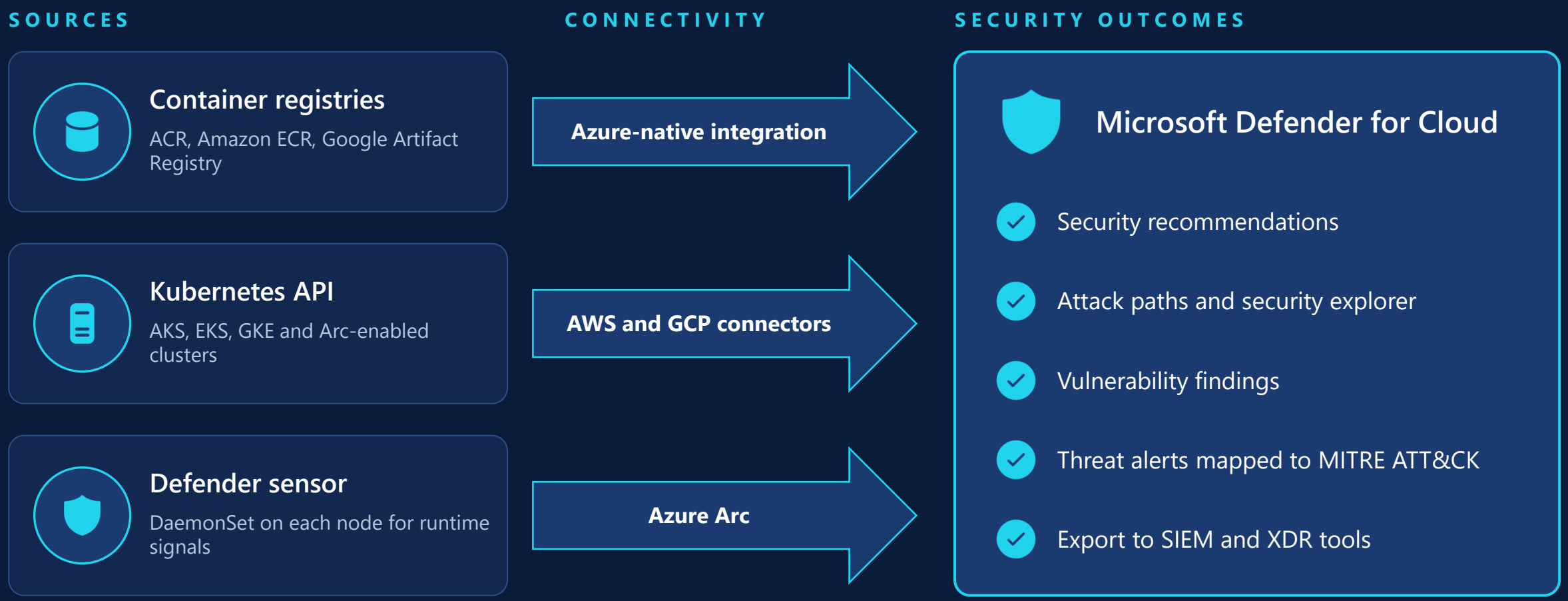
Scans trigger when images are pushed and periodically after. Results can be published back to the registry without modifying the image.



Kubernetes API connection

Defender for Cloud connects to the cluster API endpoint to discover clusters, collect configuration data and run posture and risk analysis.

Architecture: signals flow from every cloud into a single Defender for Cloud view



Deployment: onboard once, then choose how components reach your clusters

1 Onboard

Connect the AWS account, GCP project or connect clusters to Azure Arc. AKS needs no connector.

2 Enable the plan

In Environment settings, turn on the Defender for Containers plan for the environment.

3 Provision

Auto provisioning deploys the Defender sensor and Azure Policy for Kubernetes.

4 Verify coverage

Check the coverage and recommendations views to confirm every cluster is protected.

Deployment approaches

Automatic

Components deploy automatically once the plan and settings are enabled.

Manual

Turn off auto provisioning and install components yourself, including for private clusters (preview).

Mixed

Keep auto provisioning on but exclude selected AKS, EKS or GKE clusters. Not supported for Arc-only clusters.

SUMMARY

Key takeaways



Cover the whole lifecycle

Build, registry, cluster configuration and runtime each need their own control.



One platform for every cloud

AKS, EKS, GKE and Arc-enabled clusters share one portal, graph and set of alerts.



Start agentless

Defender CSPM and Defender for Containers give posture and vulnerability visibility with no cluster footprint.



Add the sensor for runtime

Deploy the Defender sensor to detect attacks in progress and map them to MITRE ATT&CK.

06

Demo

Defender for Containers in action



SOURCES

Sources and further reading

01 **Introduction to Microsoft Defender for Containers**
learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-containers-introduction

02 **Container security architecture**
learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-containers-architecture

03 **Defender for Containers deployment overview**
learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-containers-deployment-overview

04 **Deployment planning for Defender for Containers**
learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-containers-deployment-planning

05 **Containers support matrix in Defender for Cloud**
learn.microsoft.com/en-us/azure/defender-for-cloud/support-matrix-defender-for-containers

06 **Agentless container posture in Defender CSPM**
learn.microsoft.com/en-us/azure/defender-for-cloud/concept-agentless-containers

07 **Onboard agentless containers for CSPM**
learn.microsoft.com/en-us/azure/defender-for-cloud/how-to-enable-agentless-containers

08 **Zero Trust workshop: Deploy Defender sensor**
microsoft.github.io/zerotrustassessment/docs/workshop-guidance/infrastructure/INF_070

09 **Zero Trust workshop: Runtime container threat protection**
microsoft.github.io/zerotrustassessment/docs/workshop-guidance/devsecops/DS_043

Feedback

Zeskanuj kod i zostaw swoją opinię



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https://thehacksummit.com/user.html#!/lecture/THS26-5d6e/rate